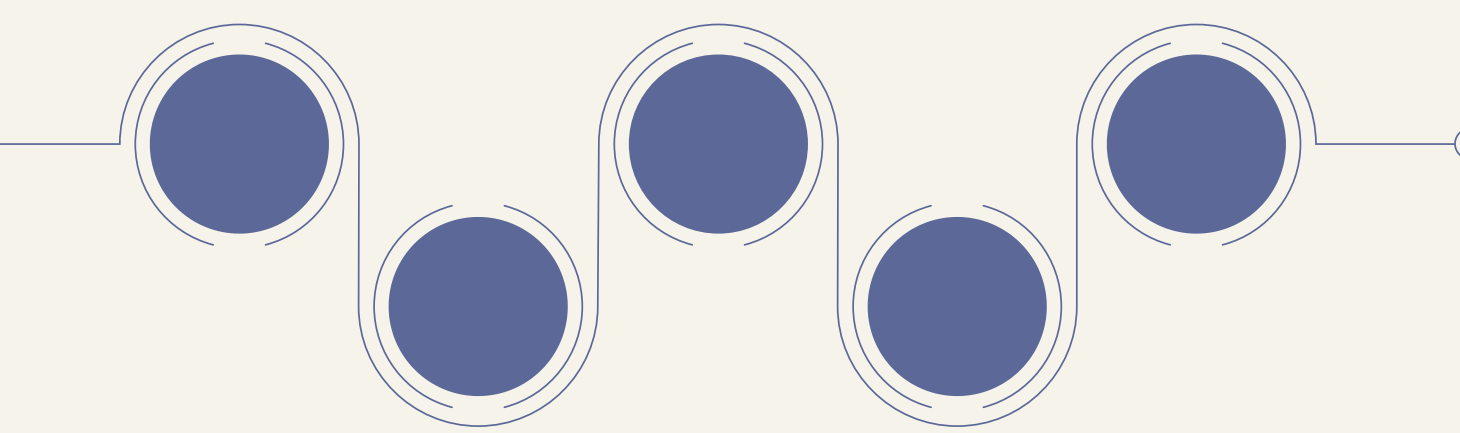




VERIFICATION

VS

VALIDATION



Presented by Team 1

Definition

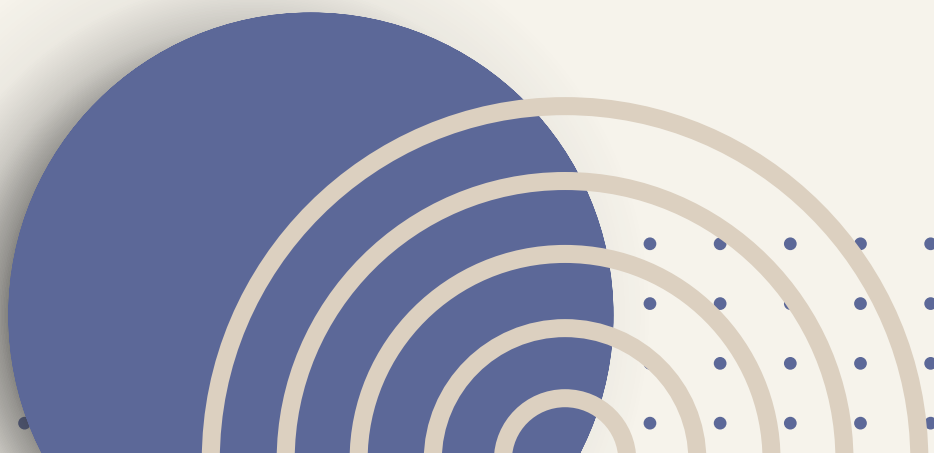


Verification

The process of ensuring that the software work products (such as requirements, design, and code) conform to specified requirements and standards. It focuses on checking correctness and consistency without necessarily executing the software.

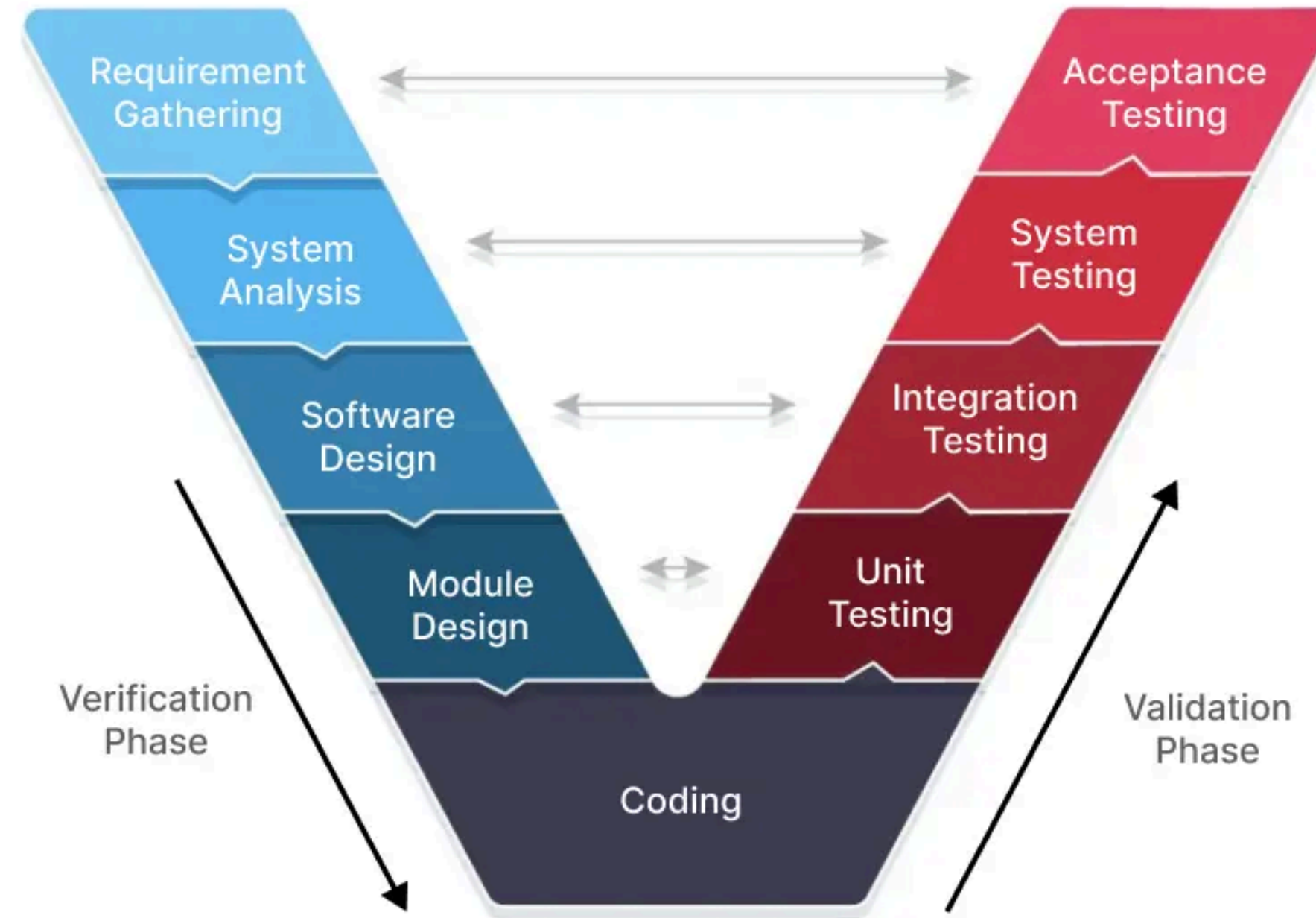
Validation

The process of evaluating the software during or after development to ensure it fulfills its intended use and meets user and stakeholder needs. It typically involves executing the software and observing its behavior in real or simulated environments.



Differences

Feature	Verification	Validation
Definition	Evaluates the intermediary work products of the software development lifecycle to check if we are on the right track.	Evaluates the final product to check whether the software meets the business needs.
Core Question	"Are we building the product right?"	"Are we building the right product?"
Focus	Checks whether the product is built as per the specified requirements and design specifications.	Determines whether the software is fit for use and satisfies the user/business needs.
Code Execution	Performed without executing the software.	Performed by executing the software.
Testing Type	Involves Static Testing techniques.	Involves Dynamic Testing techniques.
Key Activities	Reviews, Inspections, Walkthroughs, Desk-checking.	Functional testing, Regression, Systems testing, and UAT.
Responsibility	Usually performed by the development team (on documents and code).	Usually performed by the QA/Testing team and end-users.



Advantages

- **Clear Communication:** It ensures that team members understand which aspects of the software development process are focused on checking requirements (verification) and which are focused on ensuring functionality (validation).
- **Efficiency:** By clearly defining verification as checking documents and designs without executing code, and validation as testing the actual software for functionality and usability, teams avoid redundant efforts and streamline their testing processes.
- **Minimized Errors:** It reduces the chances of overlooking critical requirements or functionalities during testing, leading to a more thorough evaluation of the software's capabilities.
- **Cost Savings:** Optimizing resource allocation and focusing efforts on the right testing activities based on whether they fall under verification or validation helps in managing costs effectively.
- **Client Satisfaction:** Ensuring that software meets or exceeds client and user expectations by conducting both verification and validation processes rigorously improves overall software quality and user satisfaction.





THANK YOU

For your attention

